

MOTTEC, Switzerland

WMO: 067160

Lat: 46.1478N

Lon: 7.6242E

Elev: 5184

StdP: 12.14

Time Zone: 1.00 (EUC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	14.1	0.7	6.9	15.0	4.4	8.3	19.7	20.1	36.6	17.9	35.1	5.8	180	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.5	84.0	63.6	80.4	62.4	77.3	61.3	65.4	80.1	63.8	77.5	62.3	74.9	11.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	94.6	71.9	58.8	90.0	69.9	57.4	85.5	68.9	33.4	80.4	32.0	77.7	30.8	75.0	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.0	20.2	17.1	9.2	86.5	8.6	6.2	3.0	91.0	-2.1	94.6	-6.9	98.1	-13.2	102.6		
(5)			7.0	65.7	7.8	3.6	1.4	68.3	-3.1	70.4	-7.5	72.5	-13.2	75.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	29.9	32.3	40.4	45.8	52.5	61.1	63.7	64.0	55.5	48.4	38.4	32.4
	DBStd	14.42	8.06	9.01	9.36	8.15	8.97	8.42	6.89	7.61	7.42	7.51	8.00	7.17
	HDD50	2755	623	497	317	175	80	7	2	3	31	122	351	547
	HDD65	6758	1088	917	762	576	397	177	110	111	295	516	798	1011
	CDD50	1702	0	1	19	49	157	339	427	438	196	71	4	1
	CDD65	229	0	0	0	0	9	60	69	81	10	0	0	0
	CDH74	1495	0	0	0	2	55	433	443	536	26	0	0	0
	CDH80	347	0	0	0	0	3	106	90	148	0	0	0	0
Wind	WSAvg	5.9	4.8	5.7	6.5	6.9	7.0	6.7	6.6	6.3	5.9	5.0	4.7	4.9
Precipitation	PrecAvg	27.8	2.0	1.9	1.9	1.9	2.8	2.6	2.6	2.7	2.0	2.2	2.6	2.4
	PrecMax	36.2	5.8	8.1	4.4	3.9	6.3	5.1	6.1	5.6	6.1	6.6	7.3	6.2
	PrecMin	17.3	0.4	0.1	0.6	0.1	0.9	1.2	0.8	1.0	0.6	0.2	0.7	0.2
	PrecStd	4.5	1.4	1.8	1.0	1.0	1.7	1.1	1.0	1.1	1.3	1.5	1.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	55.1	64.9	71.2	79.5	87.2	86.9	90.6	77.3	69.7	57.3	52.5
		MCWB	40.9	42.1	47.2	51.8	59.1	64.6	63.3	63.8	61.5	55.0	45.9	42.0
	2%	DB	45.4	50.4	60.1	66.0	74.6	83.5	82.3	83.7	73.7	64.8	54.0	46.4
		MCWB	37.9	38.6	45.8	49.2	56.8	63.2	63.5	63.5	59.8	52.2	44.1	39.1
	5%	DB	42.7	46.8	56.5	61.7	70.2	79.4	78.8	79.5	69.9	62.1	50.7	43.6
		MCWB	36.8	37.5	44.1	46.9	54.5	61.6	62.5	62.4	57.3	51.1	42.5	37.7
	10%	DB	39.8	43.7	52.9	58.3	66.1	75.2	75.6	76.1	66.6	58.8	48.1	41.4
		MCWB	34.6	36.3	42.1	45.4	52.7	59.6	60.7	61.3	55.7	49.4	42.0	35.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.7	43.4	49.4	53.2	60.9	66.7	66.9	67.2	63.6	56.5	49.7	43.8
		MCDB	45.4	49.7	59.6	66.5	75.9	83.5	81.4	83.4	75.0	64.7	52.8	48.2
	2%	WB	39.5	41.1	47.2	50.3	58.0	64.9	65.1	65.0	60.6	54.6	46.8	41.8
		MCDB	44.0	47.6	57.3	63.8	71.3	80.5	78.7	79.6	70.7	63.0	50.4	45.7
	5%	WB	37.4	38.9	45.3	48.2	55.8	62.8	63.5	63.4	58.8	52.5	44.1	39.1
		MCDB	41.5	45.5	54.3	60.0	67.9	76.0	76.6	76.8	67.4	59.6	48.9	42.8
	10%	WB	35.3	36.8	43.2	46.0	53.6	60.9	61.8	61.9	57.1	50.5	42.4	36.5
		MCDB	39.1	43.0	51.9	56.7	64.7	73.3	73.5	74.2	64.6	57.0	47.1	40.2
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.5	15.4	16.9	17.1	17.9	18.3	17.5	15.2	13.7	10.4	10.0
		MCDBR	10.5	14.4	18.0	20.7	21.0	22.3	20.5	20.5	17.6	16.2	13.1	11.3
		MCWBR	6.1	8.1	9.4	10.1	10.5	9.9	9.1	9.0	8.8	8.5	7.6	7.0
	5% WB	MCDBR	10.0	12.2	16.3	19.0	18.8	19.2	18.5	18.2	14.8	13.4	9.9	8.1
		MCWBR	6.6	7.9	8.6	10.0	10.0	9.2	8.8	8.5	7.9	7.6	6.8	6.2
Clear-Sky Solar Irradiance	taub	0.239	0.266	0.313	0.358	0.377	0.382	0.378	0.365	0.341	0.314	0.273	0.238	
	taud	2.304	2.236	2.162	2.160	2.210	2.245	2.255	2.304	2.362	2.411	2.407	2.340	
	Ebn at Noon	279	290	288	284	282	280	279	279	277	269	263	266	
	Edh at Noon	24	32	40	44	44	43	42	38	33	27	22	21	
All-Sky Solar Radiation	RadAvg	471	749	1105	1395	1590	1889	1890	1643	1282	830	482	388	
	RadStd	55	100	151	189	137	163	152	132	107	92	69	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (107 neighbors)	N/A	N/A	N/A	+1.30	+0.74	+0.56	N/A	N/A	+101	+29

Nomenclature: See separate page